Fast and High-Fidelity Quantum Logic Gates for Parametrically Coupled Transmons

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INTRODUCTION

There have been significant advancements in realizing quantum computers composed of quantum bits (=qubits). However, a key challenge is the decoherence, which limits how long information can be stored in each qubit. Thus, we explore fast and high-fidelity quantum logic gates in parametrically driven transmons.

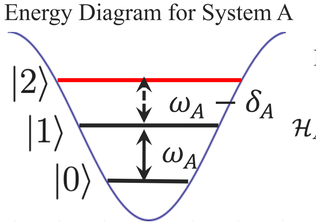
BACKGROUND

Qubits are mathematical representations of quantum states:

$$|0\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix}, \quad |2\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

A transmon is a type of superconducting charge qubit. Consider a transmon with anharmonicity δ:

Energy Diagram for System A

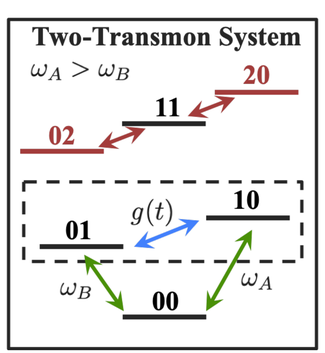


Hamiltonian for System A

$$\mathcal{H}_A = \hbar \begin{pmatrix} 0 & 0 & 0 \\ 0 & \omega_A & 0 \\ 0 & 0 & 2\omega_A - \delta_A \end{pmatrix}$$

SYSTEM OF TWO TRANSMONS

We consider the system of two coupled transmons, labelled A and B, that is parametrically driven by a pulse.



Hamiltonian for System

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{pulse}}(t)$$

$$\mathcal{H}_0 = \mathcal{H}_A \otimes I + I \otimes \mathcal{H}_B$$

$$\mathcal{H}_{\text{pulse}}(t) = g(t)\, \mathbb{X} \otimes \mathbb{X}$$

Coupling Operator

$$\mathbb{X} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

$g(t)$ is the driving pulse.

METHODOLOGY

We parametrically drive the system with a time-varying pulse, $g(t)$, for the time of T_{pulse}, activating the coupling operator:

$$g(t) = \frac{A}{2}\left(1 - \cos\left(\frac{2\pi t}{T_{\text{pulse}}}\right)\right)\cos(\omega_d t) + B \sin\left(\frac{2\pi t}{T_{\text{pulse}}}\right)\sin(\omega_d t)$$

Both the one-quadrature approach ($B = 0$) and the two-quadrature approach ($B \neq 0$) are considered. We explore pulses with drive frequencies ω_d near $\omega_A - \omega_B$. By solving the Schrödinger equation, we note that the evolution operator, U, causes the transitions between **01** and **10**. Further, we observe that the operator is locally equivalent to $\sqrt{\varphi\text{-SWAP}}$.

Schrödinger equation

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = \mathcal{H}|\psi(t)\rangle$$

$$\downarrow$$

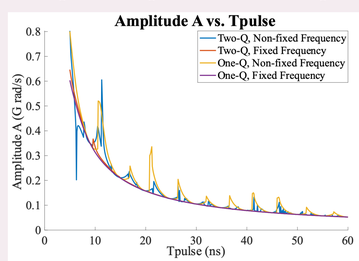
$$|\psi(T_{\text{pulse}})\rangle = U|\psi(0)\rangle$$

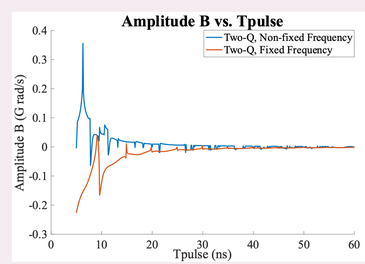
$$U \to U_{\text{local}} \sim \sqrt{\varphi\text{-SWAP}}$$

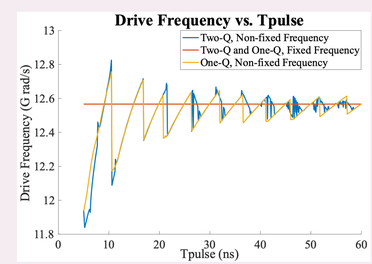
$$\sqrt{\varphi\text{-SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} & 0 \\ 0 & \frac{-i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & e^{-i\varphi} \end{pmatrix}$$

RESULTS

The optimized parameters of the pulses are for one- and two-quadrature pulses (fixed frequency is at $\omega_d = \omega_A - \omega_B$):







Using the optimized parameters, we simulate the evolution operator of the Hamiltonian from RFT without RWA, and then calculate the error.

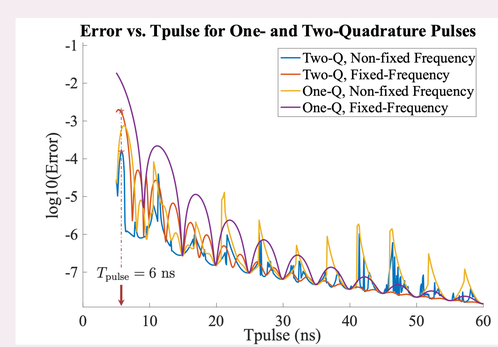
RFT with RWA: Optimize $1 - \frac{1}{4}\left|\text{trace}(\tilde{U}_{\text{local}}\sqrt{\varphi\text{-SWAP}})\right|$

Obtain optimal parameters: A, B, ω_d

$\downarrow$

RFT without RWA: Simulate using optimized A, B, ω_d

Calculate error



MODELING AND OPTIMIZATION

We impose the rotating frame transformation (RFT) and rotating wave approximation (RWA) on the Hamiltonian:

$$\mathcal{H} \sim \tilde{\mathcal{H}} = \mathcal{H}_1 + g_1(t)\mathcal{H}_2 + g_2(t)\mathcal{H}_3$$

$$g_1(t) = \frac{A}{2}\left(1 - \cos\left(\frac{2\pi t}{T_{\text{pulse}}}\right)\right), \quad g_2(t) = B \sin\left(\frac{2\pi t}{T_{\text{pulse}}}\right)$$

The evolution operator from $\tilde{\mathcal{H}}$ was numerically approximated to the second-order with small timesteps dt:

$$U \sim \tilde{U} = \prod_{t=0,dt,2dt,\ldots}^{T_{\text{pulse}}} e^{-i\mathcal{H}_1 dt/2\hbar} e^{-i(g_1(t)\mathcal{H}_2 + g_2(t)\mathcal{H}_3)dt/\hbar} e^{-i\mathcal{H}_1 dt/2\hbar}$$

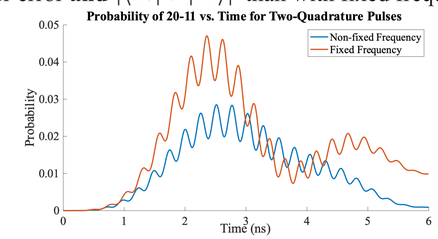
After local operations, $\tilde{U} \to \tilde{U}_{\text{local}}$, we optimize the parameters of the pulse to minimize the error between $\tilde{U}_{\text{local}}$ and $\sqrt{\varphi\text{-SWAP}}$ for various values of T_{pulse}:

$$\text{ERROR} = 1 - \frac{1}{4}\left|\text{trace}(\tilde{U}_{\text{local}}\sqrt{\varphi\text{-SWAP}})\right|$$

$$\text{FIDELITY} = 1 - \text{ERROR}$$

DISCUSSION

- Using optimized parameters for two-quadrature, non-fixed drive frequency pulses, we achieve errors between 10^{-4} and 10^{-6} for T_{pulse}'s between 5 ns and 10 ns.
- The probabilities that **11** state transitions to the **02** and **20** states, $|\langle 02|U|11\rangle|^2$ and $|\langle 20|U|11\rangle|^2$, contribute to the gate errors. For example, at $T_{\text{pulse}} = 6$ ns, two-quadrature pulses with non-fixed frequency lead to lower error **and** $|\langle 20|U|11\rangle|^2$ than with fixed frequency:



- Two applications of the $\sqrt{\varphi\text{-SWAP}}$ gates can be used to create a CNOT gate, up to a global phase shift:

[redacted]

- CNOT gates are important in the development of quantum computers, for any unitary gate can be implemented to arbitrary accuracy using CNOT and single-qubit gates.
- In conclusion, by using fast and high-fidelity $\sqrt{\varphi\text{-SWAP}}$ gates, one can achieve fast and high-fidelity CNOT gates and thus improve the accuracy of any gate.

FUTURE OUTLOOK

- Use the Derivative Removal by Adiabatic Gate (DRAG) formalism to theoretically predict optimal parameters of the pulses as analytical functions of T_{pulse}.
- Investigate locally different quantum logic gates by considering different drive frequency regions.
- Numerically model an open quantum system to explore the effects of decoherence and dephasing.
- Investigate the first-principles analysis of multi-transmon circuits.